Lesson 7 Servo Motion Speed Control and Source Code Analysis

1. PWM Servo Description

A servo is a common motor drive device used to control the position, speed, and acceleration in a mechanical system. It typically consists of a motor, a control circuit, and a feedback system.

In terms of type, servos can be categorized into analog servos, PWM (pulse width modulation) servos, and bus servos. This section mainly focuses on PWM servos.

PWM servos use pulse width modulation signals to control the angular position of the servo. In simple terms, a PWM servo controls the angular position by varying the pulse width of the signal. The range of the pulse width determines the extreme angular positions of the servo, while the period of the control signal determines the refresh frequency of the signal. By adjusting the duration of the pulse width, precise control of the servo position can be achieved.

PWM achieves a level state between 0 and 1, meaning that when PWM is output, the voltage can be any value between 0 and 3.3V/5V.

When using PWM control, it is essential to know two parameters of the object being controlled:

Its PWM frequency (corresponding to the timer cycle)

Its operating pulse width (corresponding to the PWM duty cycle)

The control signal enters the signal modulation chip from the receiver channel to obtain a DC bias voltage. Inside, a reference circuit generates a reference signal with a period of 20ms and a width of 1.5ms. The obtained DC bias

voltage is compared with the voltage of the potentiometer to produce a voltage difference output. Finally, the positive and negative outputs of the voltage difference are sent to the motor driver chip to determine the motor's forward and reverse rotation. When the motor speed is constant, the potentiometer is driven to rotate through the cascade reduction gear, ensuring the voltage difference becomes zero, and the motor stops rotating.

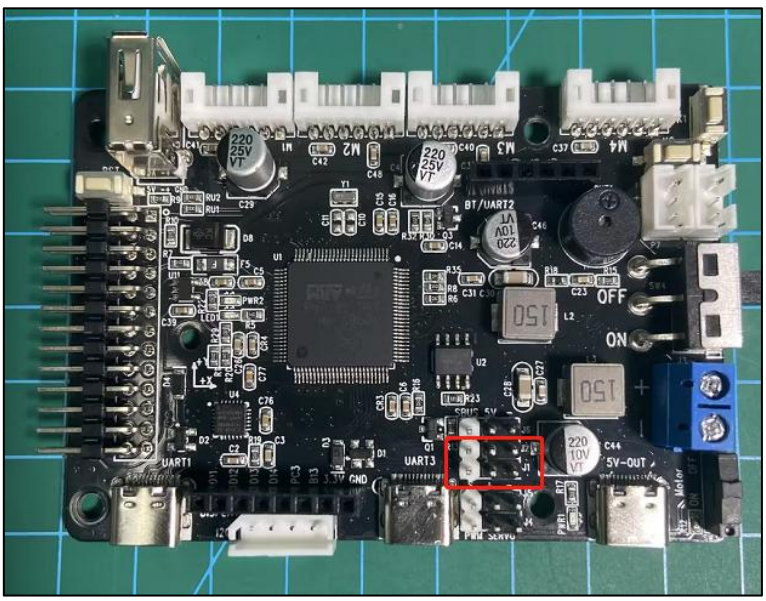
2. Wiring Method:

2.1 Required Stuffs:

1. stm32 controller
2. LD-1501MG digital servo

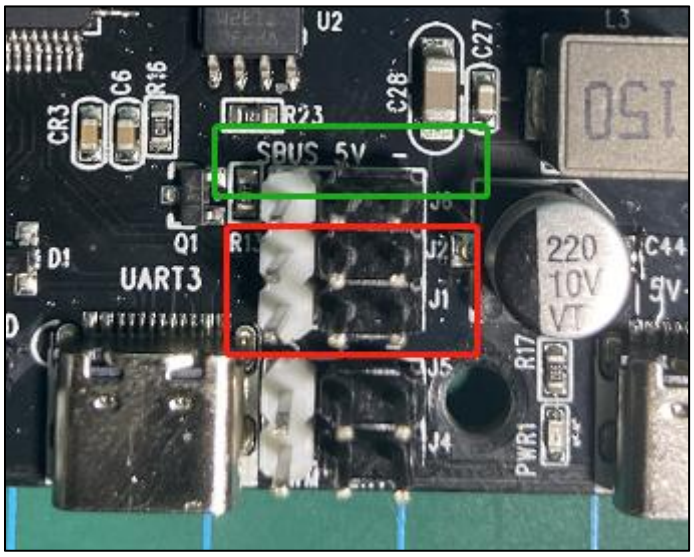
2.2 Connection Instructions

1. The diagram below shows the reserved interfaces J1 and J2 provided by the STM32 main control board for the PWM servo.



The enlarged detail image below shows the reserved connection ports J1 and J2, highlighted in the red box. There are three types of connections: signal line

(S), power supply (+5V), and ground (−), as indicated in the green box.



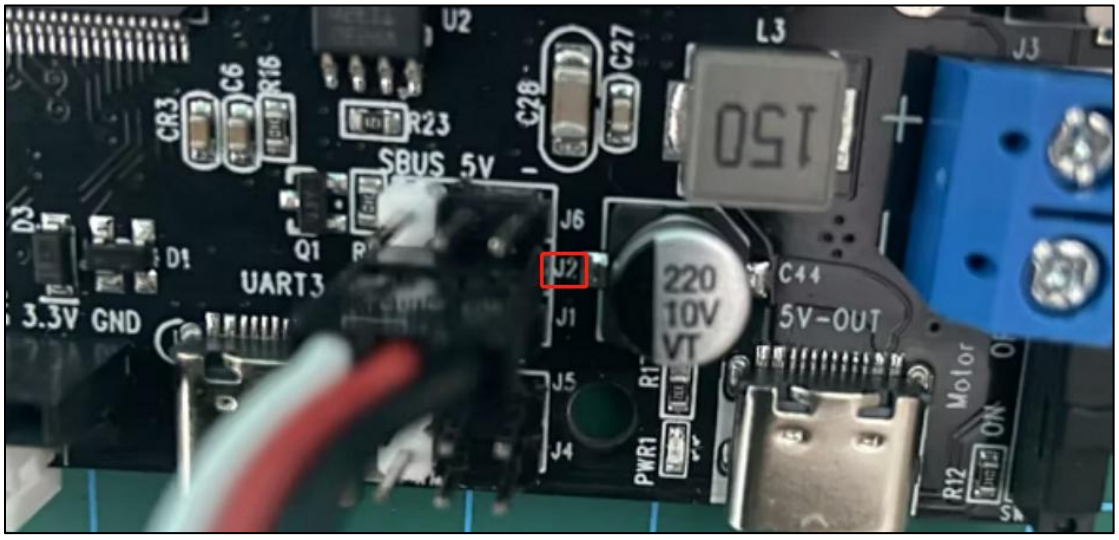
2. Connect the servo's 3-pin cable to the corresponding J2 or J1 interface, as shown below:

The connection method is as follows:

White -> Signal line (S)

Red -> Power supply (+5V)

Black -> Ground (−)



Attention: The servo motor must be connected according to the positions shown in

the document diagram. Avoid connecting it incorrectly or mistakenly to prevent burning out the motor. Our company will not take any responsibility!!!

3. Program Download

The sample program is '**rosrobotcontrollerm4_pwmServo_speed**'.

After downloading the program (refer to '2. STM32 Development Fundamentals/Lesson 3 Project Compilation and Download' for details on how to download the program), you need to press the RESET button to reset the microcontroller, which will then start executing the new program.

4. Program Outcome

After pressing reset, the servo will first initialize to 90°, then move to 0°, and finally rotate to 180° at a speed of 60 degrees per second.

5. Source Code Analysis

This program is only for speed control. For detailed PWM servo control routines, please refer to '**Lesson 6 RRC PWM Servo Control Principle and Source Code Analysis**'.

```
#define  HOLDER_MIN   500
#define  HOLDER_MAX  2500

void pwm_servos_init(void); //鑾1垫満

void app_task_entry(void *argument)
{
    uint32_t time = 3000;   //运动时间（ms）

    pwm_servos_init();  // 初始化PWM舵机控制器
    osDelay(500);      // 等待0.5秒钟，确保舵机控制器已经初始化完成
    //舵机的设置范围对应为[500 , 2500]，对应[0° ,180°]
    pwm_servo_set_position(pwm_servos[0], HOLDER_MIN, 500);     //将舵机初始化位置设置为0°（HOLDER_MIN）位置
    osDelay(500);   // 等待0.5秒钟，确保舵机控制器已经转动完成


    //设置舵机的目标值为180°（HOLDER_MAX），时间为time（ms），所以转速为 （180° - 0°）/ time
    pwm_servo_set_position(pwm_servos[0], HOLDER_MAX, time);
    osDelay(time);  //延时等待time时间，确保舵机控制器已经转动完成

    for(;;){

        osDelay(1000);  //!!!一定要延时，否则RTOS系统会出错!!!

    }

}
```

The program is located in the app.c file under /Hiwonder/System.

The **pwm_servo_set_position()** function controls the rotation of the servo. The required parameters are defined as follows:

Parameter 1: Servo object

Parameter 2: Target angle of the servo

Parameter 3: Movement time

The rotation speed can be obtained by dividing the rotation angle by the movement time. The program logic is as follows:

①Initialize the servo

②Set the servo to a known initial angle

③Delay to allow the servo to complete the movement. The delay time must be greater than or equal to the movement time of the servo, otherwise, the servo cannot reach the target position.

④Move the servo to the target angle and specify the movement time.

⑤Delay to allow the servo to complete the movement.

The servo movement speed can be calculated by dividing the known angle difference by the movement time.